KubeCon | CloudNativeCon

North America 2025

Introducing TAG Workloads Foundation: Advancing the Core of Cloud Native Execution

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Agenda

- What is a TAG?
- TAG Restructuring
- TAG Workloads Foundation
- Community Activities
- How to get involved!

What is a TAG?

What is a TAG?

Support Projects

Define and Advance Practices, Standards, Assessments

Scale contributions by the technical community

Retaining integrity increasing quality

In support of the CNCF's mission

"... within each technical domain of cloud native systems, applications, and architectures for the benefit of the community and our project adopters. This supports the CNCF's technical vision by addressing critical problems faced by adopters and contributing to a robust cloud native ecosystem."

https://github.com/cncf/toc/blob/main/governance/tag-governance.md#technical-advisory-groups-tags

TAG Goals

Strengthen the project ecosystem

Bridge between projects

Identify gaps in the CNCF Project Portfolio

Scale / Offload the TOC

Provide Technical Expertise

Educate and inform users

Help foster project maturity

CNCF Project Landscape

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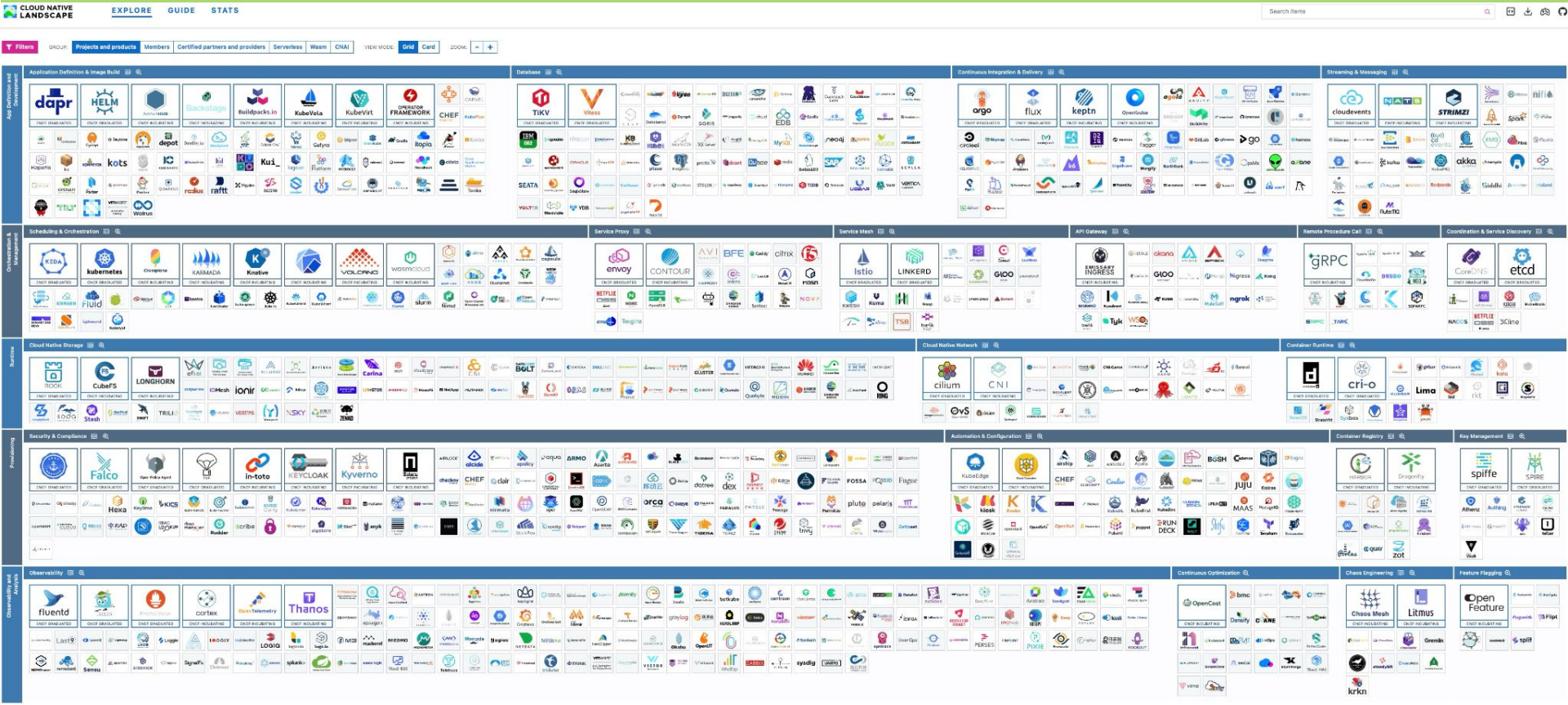

TAG Operational Resilience

- Observability
- Management
- Business Continuity
- Resource Optimization
- Cost Efficiency
- Energy
- Performance
- Troubleshooting
- Reliability
- Day 2 Ops

TAG Developer Experience

Microservices, Streaming, Messaging, API Management, Dev Frameworks

TAG Workloads Foundation

Containers, OS, Runtime, Virtual Machines, Serverless, Web Assembly, Databases, Batch, Serving, Scheduler, Orchestrator, Deployment, Dynamic Scaling, CI/CD

TAG Infrastructure

Data, Storage, Network, DNS, Compute, Service Mesh, Infrastructure-as-Code, Edge, Sovereignty, Load Balancing

TAG Security and Compliance

- Security hygiene
- Policy-as-code
- Compliance
- Auditing
- Threat Modeling
- Secure Software Supply Chain

TAG Responsibilities

Reach out to projects, engage with communities

Often the first point of contact for a project with the CNCF

Provide “Domain Technical Review”

Sponsor TAG initiatives such as whitepapers

Sponsor TAG-led Community Subprojects

Hold regular public meetings / everything in the open

Domain Technical Reviews

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What is a DTR?

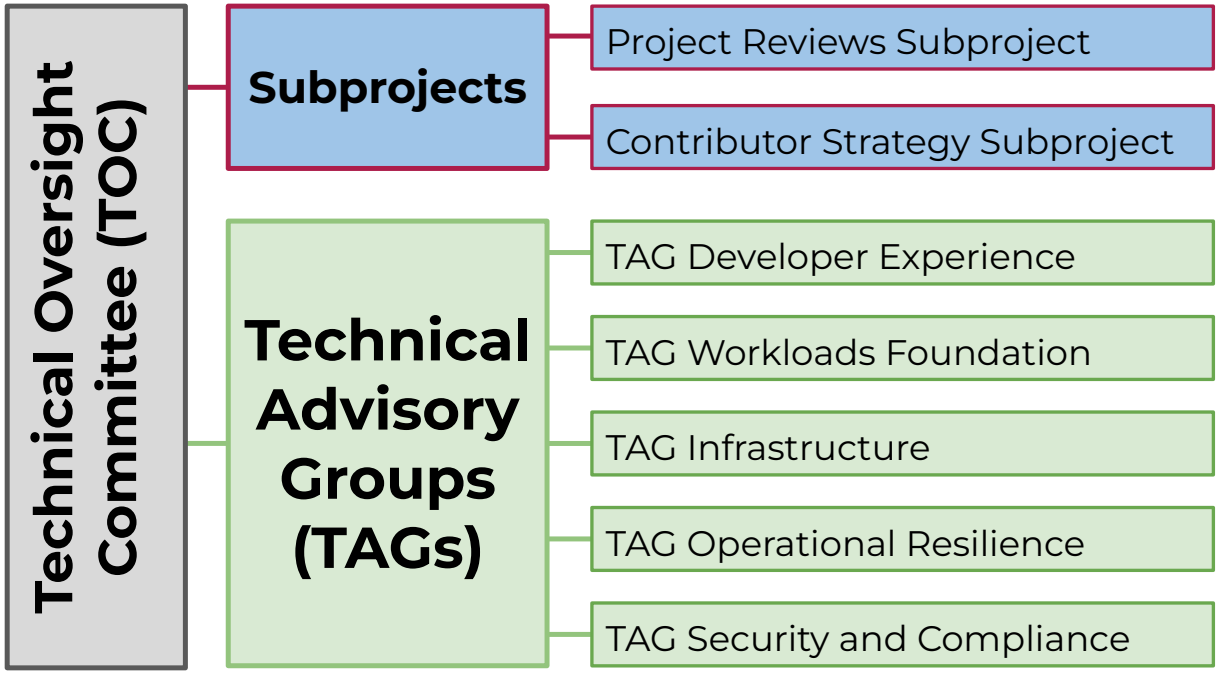
- Framework for engaging with projects to standardize across all TAGs when reviewing projects for sandbox, incubation, graduation, and providing feedback and recommendations to the TOC
- Set of General Technical Questions based on project level
- Domain specific criteria added on for TAG domain specific
- Templates for questions and feedback
- Due 2 weeks prior to TOC review to allow project to respond

General Technical Questions

- **Day 0 - Planning Phase. (Sandbox)** - design and architecture of the cloud native project.
- **Day 1 - Installation and Deployment Phase (Incubation)** - initial installation and deployment of the design
- **Day 2 - Day-to-Day Operations Phase (Graduated)** - post-deployment operations in production-ready environments to include monitoring, maintenance, auditing and troubleshooting.

Restructuring of TAGs

TAG Reboot



- Consolidated TAGs from 8 to 5
- Replace Working Groups with Subprojects and Initiatives
- Alignment with K8s working group concept
- Any previous Working Group may apply to be a new Subproject or Initiative

TAG Reboot References

TAG Reboot TOC GitHub Issue

TAG Reboot Presentation Slides

What is TAG Workloads Foundation?

Mission

To define and advance practices and standards for **fundamental cloud native workload execution environments** and their related **lifecycle management** within cloud native systems, applications, and architectures.

Scope

- Containers
- OS (Special Purpose Operating Systems)
- Runtime
- Virtual Machines
- Serverless
- Web Assembly
- Databases
- Batch
- Serving
- Scheduling
- Orchestrator
- Deployment
- Dynamic Scaling
- CI/CD

Out of Scope

- Anything not explicitly mentioned in the scope (shown in the previous slide) is out of scope.
- Any end-user directed activities and initiatives.
- Technologies that are not cloud-native (defined in CNCF Cloud Native definition)

Leadership

Chairs

- Danielle Tal (@miao0miao) (Term: 2025-07-01 - 2026-06-30)
- Paco Xu (@pacoxu) (Term: 2025-07-01 - 2027-06-30)
- Yuan Tang (@terrytangyuan) (Term: 2025-07-01 - 2027-06-30)

Tech Leads

- Alexander D. Kanevskiy (@kad) (Term: 2025-07-02 - 2026-06-30)
- Hongcai Ren (@RainbowMango) (Term: 2025-07-02 - 2027-06-30)
- Kante Yin (@kerthcet) (Term: 2025-07-02 - 2027-06-30)
- Rajas Kakodkar (@rajaskakodkar) (Term: 2025-07-02 - 2027-06-30)
- Stephen Rust (@srust) (Term: 2025-07-02 - 2026-06-30)

TOC Liaisons

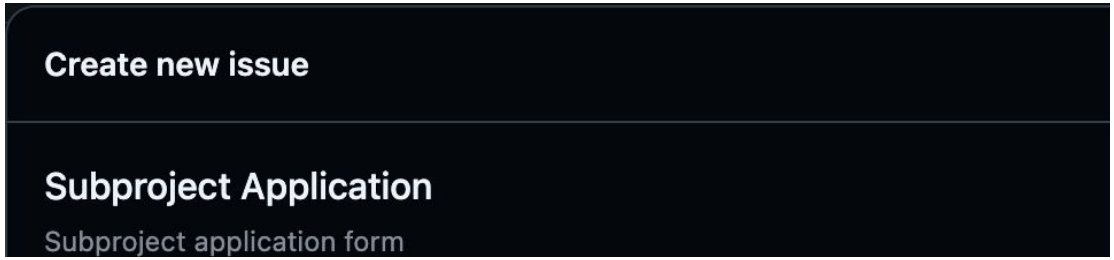
- Alex Chircop (@chira001)
- Kevin Wang (@kevin-wangzefeng)

Community Activities

Subprojects

Propose a Subproject

- Open a new issue to propose an subproject! https://github.com/cncf/toc/issues

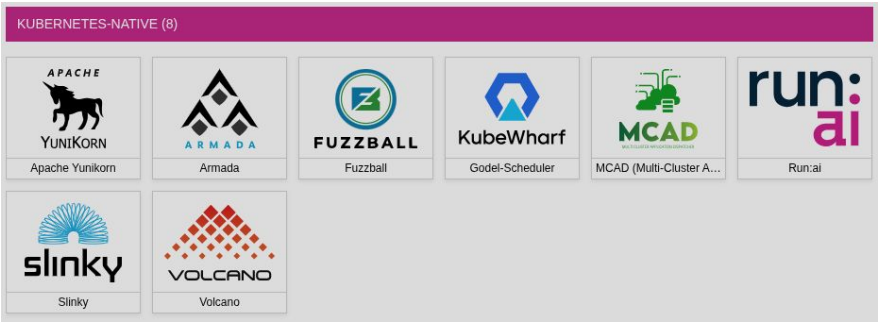


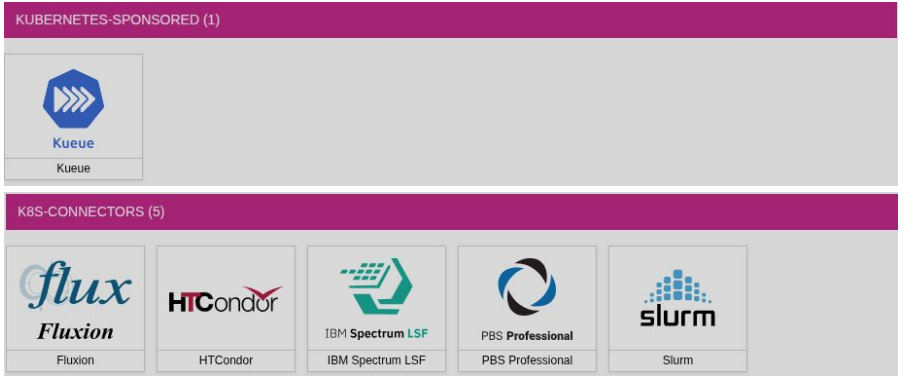
Batch Systems Subproject

Batch Systems Subproject

"To define and advance practices and standards for fundamental cloud native workload execution environments and their related lifecycle management within cloud native systems, applications, and architectures."

Landscape:
https://tag-runtime.cncf.io/wgs/bsi/landscape/





Batch Systems Subproject

Meetings biweekly Tuesdays at 8AM PST, next meeting December 2nd

Current Initiatives:

- Scheduling for Cloud Native AI
  - To enumerate and educate the various challenges and opportunities regarding optimizing resource allocation for Cloud Native Artificial Intelligence (CNAI) workloads
- Data Locality Initiative
  - IO is hard. An ongoing discussion on how to deal with the fact that some workloads have a lot of data and ways to deal with that when relocating across clusters.
- Batch Definitions for a Cloud-Native+HPC Universe
  - Definitions are currently confusing. The Batch working group is writing a doc to firm up these definitions to create a common language.

Initiatives

Initiative: Cloud Native AI Scheduling Challenges Whitepaper

- TAG Workloads Foundation stewarding this initiative under the Batch Systems Subproject
- https://github.com/cncf/toc/issues/1641

- Please join help us drive this forward! Need to refine before approval.

"This paper aims to enumerate and educate the various challenges and opportunities regarding optimizing resource allocation (aka scheduling) for Cloud Native Artificial Intelligence (CNAI) workloads. Cloud Native allows easy scaling of resources, making it ideal for AI workloads of two types: training and inference."

[PUBLIC] Challenges in Scheduling Cloud Native AI Workloads Whitepaper

Initiative: Cloud-Native Foundations for Distributed Agentic Systems

- TAG Workloads Foundation stewarding this initiative
- https://github.com/cncf/toc/issues/1746

- Approved. Looking for contributors!

“Formalise principles, reference patterns and ecosystem strategy for running massively distributed systems of collaborating AI agents on Kubernetes and other cloud-native substrates.”

Other Initiatives

Roughly 30 other initiatives are currently underway

- Current Open Initiatives

Example Initiatives:

- AI Security white papers
- AI Sustainability - Challenges & Opportunities
- Business Continuity: white paper and best practices
- Foundations for Distributed Agentic Systems
- Community Knowledge Graph
- Compliance Use Cases White Paper
- Data Storage in AI
- Improving Developer Experience for Secure Coding
- MCP Server Standards Whitepaper
- Observability Query Language Standardization
- Security Controls Catalog Refresh
- Storage Landscape V3
- Supply Chain Security Insights

Propose an Initiative

- Open a new issue to propose an initiative! https://github.com/cncf/toc/issues

Create new issue

Initative

Initiative request form

Community Groups

Projects

Graduated Projects

Graduation

Maturity and Sustainability

Thriving Adoption

Healthy Governance Structure

Strong Diverse Community

High Standards for Security and Compliance

“This includes adopting the CNCF Code of Conduct, maintaining the Core Infrastructure Initiative (CII) Best Practices Badge, completing security reviews, and proving a stable committer process with multi-organization backing.”

https://github.com/cncf/toc/blob/main/.github/ISSUE_TEMPLATE/template-graduation-application.md

Continuous Integration and Delivery

2022-11-30

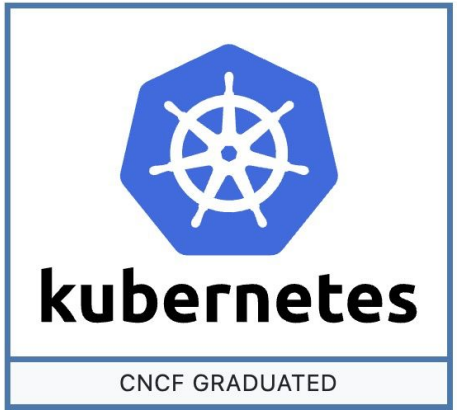
2022-12-06

Database

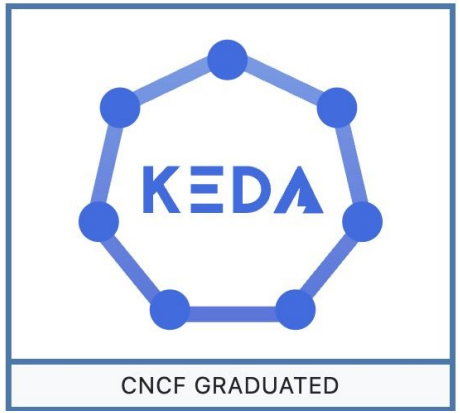
2019-11-05

2020-09-02

Scheduling and Orchestration



2018-03-06



2023-08-22

2025-09-11

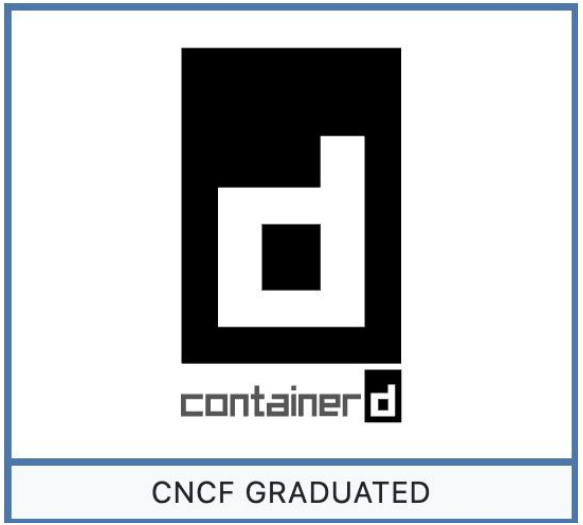
2025-10-28

Container Registry

2020-06-15

2025-10-28

Container Runtime



2019-02-28

2023-07-19

Incubating Projects

Incubation

Significant Production Adoption

Publicly documented list of adopters

Strong Vendor-Neutral Governance

Clear Maintainer Lifecycle

“Projects also need to demonstrate sustainability, a healthy contribution flow, and adopt the CNCF Code of Conduct.“

https://github.com/cncf/toc/blob/main/.github/ISSUE_TEMPLATE/template-incubation-application.md

Scheduling and Orchestration

KARMADA	Kubeflow	VOLCANO	OpenKruise
CNCF INCUBATING	CNCF INCUBATING	CNCF INCUBATING	CNCF INCUBATING
2023-12-12	2023-07-25	2022-03-21	2023-03-02

OS / Container Runtime / Virtual Machines

KubeVirt

CNCF INCUBATING

2022-04-19

Lima

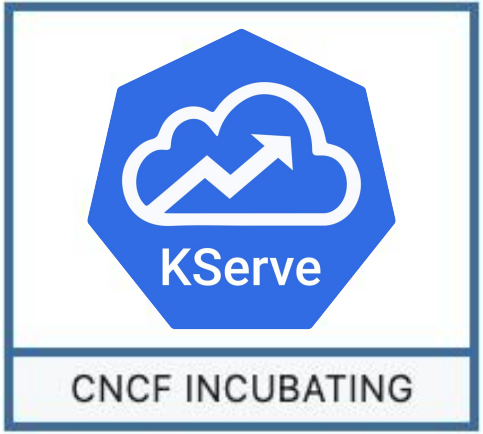
CNCF INCUBATING

2025-10-14

CNCF INCUBATING

2024-08-02

Serving



2025-09-29

WebAssembly

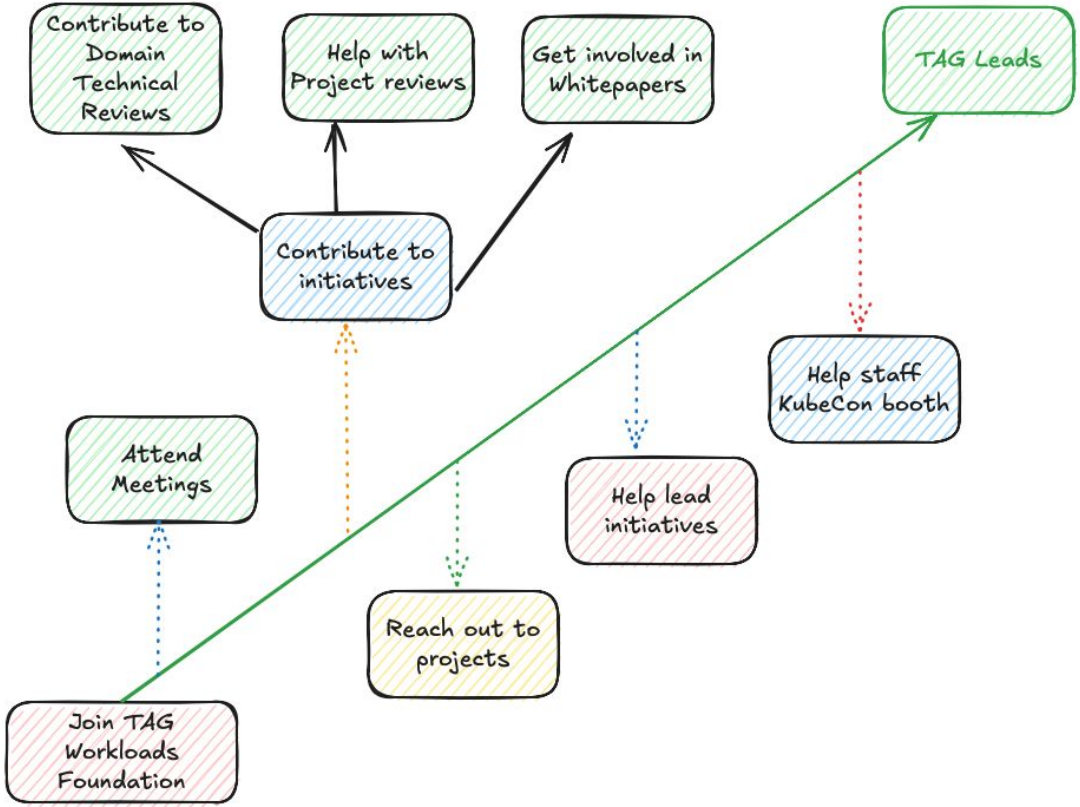
wasmcloud

CNCF INCUBATING

2024-11-08

Getting Involved

Contributor Ladder



How to get involved?

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- TAG Community Meeting:
  - First Wednesday of each month at 13:00 UTC
  - Agenda
- Slack Channel: #tag-workloads-foundation
- Mailing List
- TAG Initiatives

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